D^{3}Former: Dual-Decoder Dual-Transformer Reconstruction Network for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) aims to identify and locate anomalous targets in hyperspectral image (HSI) data that are significantly different from the surrounding background or environment. Most existing deep learning (DL) methods for HAD utilize reconstruction errors by reconstructing the background, but they often result in equivalent reconstruction of both background and anomalies, failing to effectively separate them. To address this issue, we propose a novel dual-decoder dual-transformer reconstruction network (D^{3}Former). Specifically, we introduce a new background feature purification decoding task, which focuses attention on the background direction during image reconstruction. In addition, we develop an anomaly stripping auxiliary branch that uses an autoencoder (AE) and dynamic large convolution kernels to mask anomalies and learn local background information, further diminishing the impact of anomalies on the reconstruction process. Furthermore, we design a new cross-guided multihead self-attention mechanism that integrates multiscale background attributes globally to suppress anomalous pixels and guide the reconstruction of a pure background. Finally, we adopt a synergistic optimization (SYO) loss to achieve twofold suppression of potential anomalies in both spatial–spectral and frequency domains, resulting in more accurate detection. Extensive experiments on multiple datasets demonstrate that our method outperforms existing state-of-the-art approaches in detection performance. The codes are available at https://github.com/YKYANG01/TGRS-D3Former.**

***Index Terms*—Autoencoder (AE), dual-decoder, hyperspectral anomaly detection (HAD), image reconstruction, transformer.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are of significant importance in remote sensing due to their ability to identify fine spectral features of ground objects, particularly excelling in target detection and localization [1], [2]. Hyperspectral anomaly detection (HAD) involves identifying anomalous targets with spectral features that are significantly different from the background in the absence of prior information. HAD finds wide applications in fields such as Earth sciences [3], remote sensing, agriculture, military, and environmental monitoring [4]. Anomalies typically refer to targets that are significantly different from the background in the spectral domain and occupy a very small area. In HAD tasks, the lack of prior knowledge about targets and backgrounds makes the separation of anomalies from the background be a key challenge [5], [6]. To address this, various methods have been proposed to model the background using different strategies to detect anomalous targets that deviate from background patterns [7], [8]. Existing HAD methods are mainly categorized into three types: statistical theory-based methods, representation learning-based methods, and deep learning (DL)-based methods.

A. Statistical Theory-Based Methods

Statistical theory-based HAD methods typically model the background using the statistical characteristics of HSI. Among these, the Reed–Xiaoli (RX) detector is pioneering, assuming a multivariate Gaussian distribution for the background and detecting anomalies via the calculation of the Mahalanobis distance between the global pixels and the background, thereby acting as a global RX (GRX) detector [9]. To emphasize local background features, local RX (LRX) [10] computes the covariance matrix using a sliding window. Advanced methods, such as the kernel isolation forest detector (KIFD) [11], use the isolation forest algorithm to model local backgrounds. Tao et al. [12] enhance anomaly detection (AD) with score Fourier entropy (FrFE). The two-step generalized likelihood ratio test (2S-GLRT) [13] employs a sliding double-window technique to improve the accuracy of multipixel AD through a two-step process.

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B. Representation Learning-Based Methods

Most statistical theory-based AD methods assume that the background conforms to a certain statistical distribution model. However, due to the complexity of HSI backgrounds and the interference from various factors, these assumptions are difficult to meet in practice. In contrast, representation-based [14], [15] HAD methods reconstruct background pixels using a dictionary, where anomalous pixels fail to reconstruct effectively, with reconstruction errors used for detection. Representation learning encompass sparse representation (SR) [16], collaborative representation (CR) [17], [18], and low-rank representation (LRR) [19], [20]. SR assumes background samples are represented by a few atoms, while anomalies cannot be done. CR emphasizes collaborative relationships among dictionary atoms, representing a pixel as a linear combination of its neighbors [21]. Improved CR models include the CR-based detector (CRD) [22] and relaxed CR detector (RCRD) [23]. LRR captures global structures by decomposing HSI into a low-rank background and a sparse anomaly matrix. Based on LRR theory, Li et al. [24] combined low-rank and sparse decomposition with a mixture of Gaussian (LSDM-MoG) using Manhattan distance to reduce noise effects. To utilize local geometric structural information from HSI, Feng et al. [25] proposed the local spatial constraint and total variation (LSCTV) method. In [26], a learnable background endmember with subspace representation (LEBSR) was developed to optimize background dictionary construction. He et al. [27] proposed the tensor low-rank approximation (TLRA-MSL) model that combines spatial and spectral information for HAD.

C. DL-Based Methods

In recent years, DL [28] has made significant progress in the fields of computer vision and remote sensing [29], [30]. DL-based HAD methods enhance the ability to recognize backgrounds and anomalous targets by extracting deep features. Common methods are divided into the similarity measurement-based models and the reconstruction-based models.

Similarity measurement models use a convolutional neural network (CNN) as the core framework, with pixel pairs selected from labeled HSI during training to learn pixel similarity for AD. Li et al. [31] proposed the CNN-based detection (CNND) method outputs anomaly scores by comparing the differences between pixel pairs. Rao et al. [32] designed a transferable Siamese network (TSN) that converts pixel similarity into anomaly probabilities. However, these methods rely heavily on the quantity and quality of labeled data, resulting in limited transferability.

Relative to this, reconstruction-based unsupervised DL techniques have demonstrated their effectiveness and achieved promising results in HAD tasks. Autoencoder (AE) [33] and generative adversarial networks (GANs) [34] are among the typically used network architectures. Jiang et al. [35] proposed a weakly supervised GANs based on spectral constraints to enhance the discriminative capability for anomalous targets. To improve the network's robustness in handling anomalies and noise, some HAD methods consider the noise model in AE, such as a fully convolutional AE (CAE, Auto-AD) [36], and robust graph AE (RGAE) [37]. The guided AE (GAED) [38] enhances the representation capability of the background. In order to better refine background recovery, SMCNet [39], KACNet [40], and some methods incorporate masking strategies and clustering concepts have been proposed. Recently, self-supervised learning (SSL) AD techniques have also garnered attention. SSL designs self-supervised tasks in HAD by leveraging techniques such as contrastive learning and masked reconstruction to enhance feature representation and generalization ability. Combined with deep networks, it constructs an efficient detection framework, providing an effective label-free solution for HAD. In [41], a self-supervised anomaly prior (SAP) network redefines the optimization criterion for anomalies in spatial structure. Sun et al. [42] proposed a contrastive self-supervised background reconstruction (CSSBR) approach to improve the model's transferability. Other methods, such as NL2Net [43] and ACRL [44], enhance surrounding background awareness by coupling nonlocal and local features. Liu et al. [45] proposed a self-supervised multiscale network (MSNet) that combines multiscale feature extraction to address the issue of reconstructing anomalies of different sizes. Approaches including PDBSNet [46] and DirectNet [47] utilize blind spot networks to reduce the feature representation of anomalies.

D. Motivation and Contributions

Although the established DL-based HAD methods have made significant progress, there are still some challenging issues that need to be addressed.

1) Most existing reconstruction-based HAD networks use a single encoding–decoding structure to extract the latent representation of the background. However, this limits the depth and diversity of feature extraction, and the single encoding–decoding path also restricts the generation of robust feature maps.
2) Existing background reconstruction-based HAD networks apply constraints on features only at the pixel level, which makes it easy to reconstruct anomalies in complex scenarios. They lack an active suppression strategy for anomalies in the background reconstruction.
3) There is a lack of constraints on anomalies in the frequency domain. In this case, the network learns to recover information from all frequencies, resulting in high-frequency anomaly features being unconsciously restored, which lowers the purification of the background.

To address these challenges, transformer [48], [49] has been used in HAD due to its powerful capability for modeling global dependencies [50], [51]. However, existing methods still have shortcomings in suppressing potential anomalies and are unable to effectively adapt to the requirements of AD.

Therefore, the core issue of HAD lies in how to effectively reconstruct background pixels through a specific feature encoding–decoding mechanism, fully utilizing the spatial correlations of the background area while eliminating the

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mapping contamination caused by anomalies. To address this issue, this article proposes an end-to-end background reconstruction method based on a dual-decoder dual-transformer reconstruction network (D^{3}Former). The newly designed dual-decoder structure can dynamically adjust its focus on the background and anomalies, fully leveraging the extracted global–local diversified features. It maps the uniform noise space into the HSI space to learn the spatial dependencies of the background and reduce the visibility of anomalous targets. To further prevent the reconstruction of anomalous pixels, an anomaly removal auxiliary branch is designed, which combines a CAE with an anomaly masking attention mechanism to effectively eliminate the significance of anomalous targets. This branch establishes an explicit suppression pathway for the anomaly's spatial–spectral structure. Notably, our proposed background reconstruction architecture features a unique cross-guided attention design that enhances the statistical dependencies of the background while effectively mitigating the spatial dependencies related to anomalous pixels. Different self-attention mechanisms cross-capture local background features, improving background quality from the perspective of structural feature complementarity. Furthermore, to improve the model's reconstruction performance, we introduce a synergistic optimization (SYO) loss function. By utilizing a frequency-domain loss term, it compensates for the limitation of reconstruction-based HAD methods that only suppress anomalies in the spatial domain. This enhancement penalizes anomalous targets more effectively in both the spatial–spectral domain and the frequency domain through background-prior mapping with weighting, allowing the model to better separate anomalies from the background.

In summary, the novelty and main contributions of this article are emphasized as follows.

1) To focus feature attention on the background during reconstruction, we propose an end-to-end background reconstruction method based on D^{3}Former. By incorporating a new background purification decoding task to prevent the network from learning features of anomalous pixels. The dual-decoder framework provides a combined bridge for different features, enabling the complementary diversity of features, which makes the generated background more robust.
2) To address the issue of anomaly contamination, we develop an anomaly removal and masking auxiliary branch (ARMAB) that utilizes AE and dynamic large convolution kernels to blur anomalous targets, supplementing a comprehensive spatial–spectral background information flow. In addition, we designed an innovative cross-guided multihead attention mechanism that globally integrates multiscale background attributes, ensuring that anomalies are not reconstructed.
3) To improve the quality of "anomaly-free" background reconstruction, we employ an SYO loss function. By introducing a new frequency-domain loss, we suppress potential anomalous components in both spatial and frequency domains. The interaction between the domains can also help preserve the texture details of the background.

The remainder of this article is organized as follows. Section II outlines the general architecture of our network and elaborates on each component. In Section III, we validate the effectiveness of our method through experiments. Finally, Section IV presents the conclusion.

II. Methodology

In this section, we will provide a comprehensive presentation of our D^{3}Former. Section II-A introduces the overall structural framework of our network. Following that, Sections II-B–II-E delve into several key components of our network, including dual-decoder, ARMAB, cross-guided transformer module (CGTM), SYO loss function, background feature fusion module (BFFM) and detection phase.

A. Overall Structure

The overall framework of the proposed D^{3}Former is shown in Fig. 1. During inference, the testing HSI is input into the fine feature (FF) encoder–decoder network, while uniform noise is fed into the lower layer background purification decoder (BPD) to simulate anomaly masking and reduce the model's visibility to anomalous targets. This dual-decoder structure facilitates hierarchical background feature extraction, improving the detection capability for diverse anomalies and backgrounds. In the upper layer, the Swin transformer [52], [53] module (SWTM) is employed as the feature extractor to capture fine background features. The FF encoder features and the BPD outputs are passed to the CGTM, explicitly guiding attention toward background semantic information and reducing the weight of anomalous pixels. The processed features are then integrated with the upper layer FF decoder outputs using the BFFM to further enhance background reconstruction. In addition, an auxiliary branch is designed for anomaly separation, where anomalies features are extracted and masked as supplementary inputs for the fused feature maps at each stage, thereby improving the model's anomaly suppression capability. Finally, the model is iteratively optimized using an SYO loss function, ensuring the generation of accurate and realistic pure background images across different decoding stages.

B. Dual-Decoder Structure for Background Reconstruction

Conventional U-shaped networks (UNet) use a single encoder–decoder connection, which results in significant information loss when processing complex HSI. Increasing network depth by stacking convolutional layers can improve feature extraction, but it also inevitably leads to the reconstruction of anomalies during background generation, reducing detection performance. To overcome these challenges, we designed a dual-decoder structure that facilitates feature sharing between layers and extracts a precise background without anomalies. BPD provides a combinatorial bridge for the diverse, nonanomalous background features, enhancing the robustness and detail richness of the reconstructed background. In BPD, all stages of background information, along with methods for anomaly suppression, are fully utilized, ensuring that the anomalies in the reconstruction background are suppressed as

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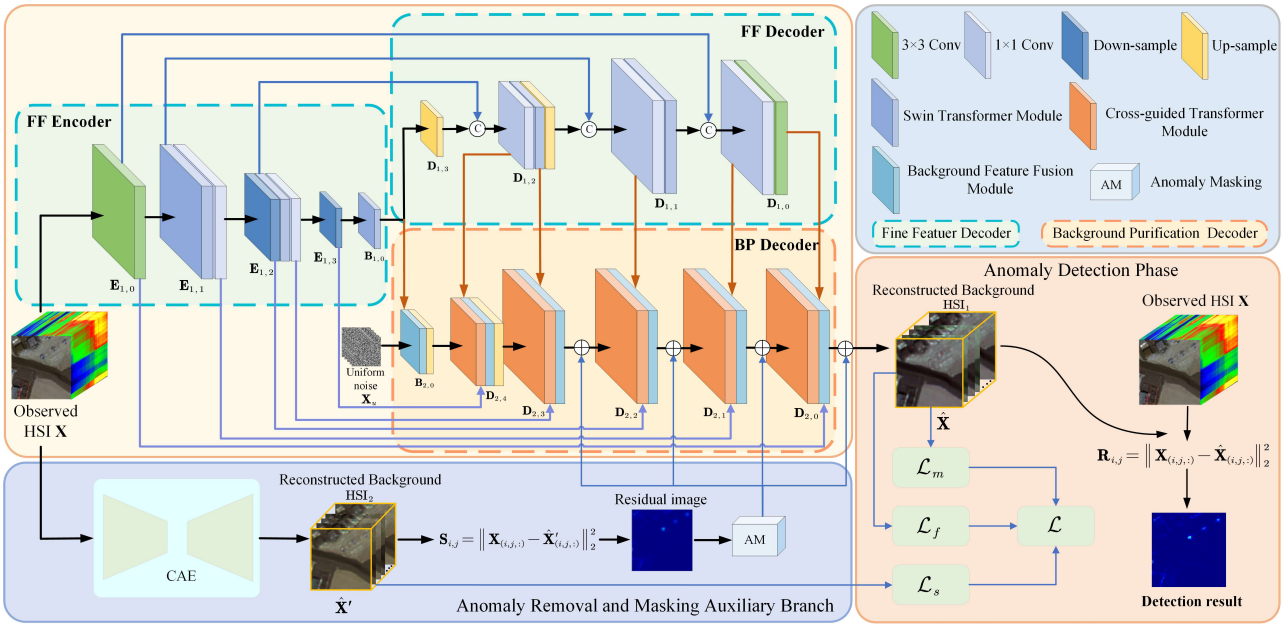


Fig. 1. D^{3}Former for HAD. The observed HSI first undergoes the FF encoding and decoding process, and the uniform noise is fed into the lower layer BPD to simulate anomaly-free input. CGTM enhances feature representation in background regions, while ARMAB reduces the weight of anomalous pixels. During the detection stage, BFFM performs cross-stage feature fusion of background information from different layers to obtain the reconstructed "anomaly-free" background HSI$_1$, and the final detection result is acquired via the difference between the "anomaly-free" background HSI$_1$ and the original HSI.

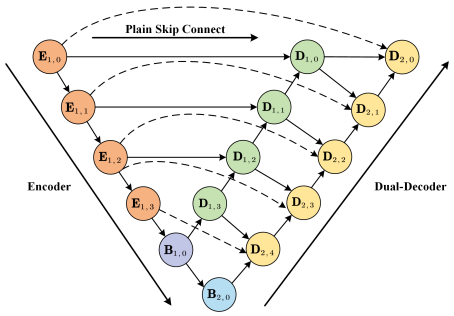


Fig. 2. Illustration of our dual-decoder structure.

much as possible. This significantly improves the quality of background reconstruction and detection accuracy.

Fig. 2 illustrates our dual-decoder structure based on UNet design, featuring two parallel decoding subnetworks. To begin with, we denote an HSI as $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, H, W, and C represent height, width, and spectral bands, respectively. The encoder receives the original HSI as input and uses a 3×3 convolutional layer to change the number of channels from C to N. The feature map then goes through two SWTMs, followed by a 1×1 convolutional layer and a downsample layer for shallow feature extraction. The downsample layer consists of a 2×2 stride convolutional layer, which reduces the spatial resolution of the feature map by half while doubling the number of channels. The sizes of the feature maps for different nodes become $\mathbf{E}_{1,2} \in \mathbb{R}^{H/2\times W/2\times 2N}$ and $\mathbf{E}_{1,3} \in \mathbb{R}^{H/4\times W/4\times 4N}$, where subscript $(\cdot)_{i,j}$ represents the jth feature extraction block of the ith encoder or decoder. The bottleneck of the model is a single SWTM, which further extracts features while keeping the shape of the feature map $\mathbf{B}_{1,0} \in \mathbb{R}^{H/4\times W/4\times 4N}$ unchanged. Then, symmetrically with the encoder, the upper decoder uses two 2×2 stride deconvolutional layers, a 1×1 convolutional layer, and an SWTM for deep feature map representation learning. After decoding, the size of the feature map returns to $H \times W \times C$.

The BPD uses the mapped input noise $\mathbf{X}_u \in \mathbb{R}^{H/4\times W/4\times 4N}$, after passing through the bottleneck BFFM and upsample layer to $\mathbf{B}_{2,0} \in \mathbb{R}^{H/2\times W/2\times 2N}$. Then, through skip connections, the feature maps from the encoding and decoding stages of the upper layer are passed to the CGTM and BFFM for feature interaction, allowing the BPD to learn more contextual information. After this stage, the final feature map generates a real background $\hat{\mathbf{X}} \in \mathbb{R}^{H\times W\times C}$.

C. Anomaly Removal and Masking Auxiliary Branch

In HAD, traditional deep models may lack sufficient information to learn the differences between anomalies and the background. As a result, they attempt to reconstruct every part of the input, including the anomalies. To enhance the differences between the background and anomaly target features and to encourage the model not to focus on the anomalies, we propose to construct an ARMAB, as shown in Fig. 3, which includes the following several components.

1) Background Anomaly Separation: The previous stage of the entire branch completes background reconstruction by using a CAE to capture local features, providing a flow of local feature information about the background from a different structural perspective. In parallel, during the anomaly separation process, the CAE preserves the overall details of the background and further suppresses the reconstruction of anomalies by using large kernel convolutions to reduce the attention on anomalous pixels. The local feature information

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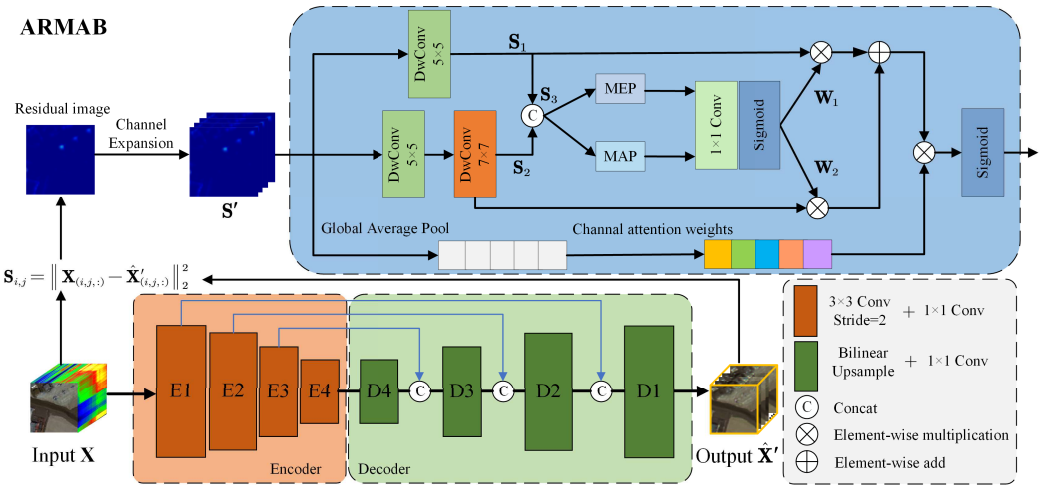


Fig. 3. Illustration of ARMAB.

flow enhances the local–global interaction from different structural perspectives, providing supplementary information for fine pure background reconstruction. The encoding block consists of a 3×3 convolution (stride = 2) and a 1×1 convolution (with batch normalization and LeakyReLU). The input of the ith encoding block can be represented as

$$\mathbf{E}_i = \mathrm{Conv}_{3\times3}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{E}_{i-1}\right)\right) \tag{1}$$

where $0 < i < 5$. The decoding block consists of an upsample layer and a 1×1 convolution (with batch normalization and LeakyReLU). Combined with skip connections, it preserves shallow features. The input of the decoding process can be represented as

$$\mathbf{D}_i = \mathrm{Concat}\left[\mathbf{E}_{i-1},\ \mathrm{Conv}_{1\times1}\left(\mathrm{Upsample}\left(\mathbf{D}_{i-1}\right)\right)\right]. \tag{2}$$

The independent path of the CAE provides multiple reconstruction perspectives and distinguishes the background from anomalies. As training progress, this subtraction approach increasingly highlights the anomaly regions, removing some redundant background information while preserving the overall background detail. In this way, our designed anomaly masking mechanism utilizes the large kernel convolutions in a feature space that has been preliminarily stripped and dominated by the background, rather than struggling to identify anomalous targets in the original features. Based on the separated feature map, subsequent large kernel convolutions can effectively blur and suppress the residual anomaly components. The calculation method for the entire residual result $\mathbf{S}$ is as follows:

$$\mathbf{S}_{i,j} = \left\|\mathbf{X}_{(i,j,:)} - \hat{\mathbf{X}}'_{(i,j,:)}\right\|_2^2 \tag{3}$$

where $(\cdot)_{i,j,:}$ is the spectral vector at position (i, j).

2) Anomaly Feature Masking: After obtaining the separated feature map $\mathbf{S}$, $\mathbf{S} \in \mathbb{R}^{H\times W}$ is replicated along the spectral dimension using channel expansion operation, transforming it into tensor $\mathbf{S}' \in \mathbb{R}^{H\times W\times C}$. Since anomalies are small and isolated pixels, which occupy only a small portion of the space. We introduce an anomaly masking mechanism to eliminate the feature space mapping of anomaly information. In the anomaly masking phase, we divide it into three parts to perform hierarchical suppression of anomalies. First, we use a separate large convolution kernel along with two sets of cascaded large convolution kernels in parallel. Through hierarchical extraction, anomalies are flattened by the convolution calculations of the surrounding background pixels. This reduces the strong distinguishability of the anomalies, achieving a blurring effect. The large convolution kernels provide a wider receptive field, focusing on larger background areas, thereby "diluting" and dispersing the anomalous information within the feature maps $\mathbf{S}_3$

$$\mathbf{S}_3 = \mathrm{Concat}\left[\mathbf{W}^{\mathrm{Dw}}_{5\times5}\left(\mathbf{S}'\right),\ \mathbf{W}^{\mathrm{Dw}}_{7\times7}\left(\mathbf{W}^{\mathrm{Dw}}_{5\times5}\left(\mathbf{S}'\right)\right)\right] \tag{4}$$

where $\mathbf{W}^{\mathrm{Dw}}_{k\times k}$ represents 5×5 depth-wise convolution (DwConv) and 7×7 DwConv.

Then, the obtained feature maps are passed through a combination of mean pooling (MEP) and max pooling (MAP) for processing. This process allows MEP to reduce the relative value of anomalies and focus on the globally consistent background. MAP selects local background information, ensuring that anomalies are no longer the focal point of the global information. Following this, a 1×1 convolution and an activation function are applied to obtain the background selections $\mathbf{W}_1$ and $\mathbf{W}_2$

$$[\mathbf{W}_1, \mathbf{W}_2] = \mathrm{Sigmoid}\left(\mathrm{Conv}_{1\times1}\left[\mathrm{MAE}\left(\mathbf{S}_3\right), \mathrm{MAX}\left(\mathbf{S}_3\right)\right]\right). \tag{5}$$

The processed information is passed into two parallel convolutional structures and combined using element-wise addition to reduce the prominence of anomalies in the fused feature map. A channel attention mechanism is then applied to dynamically reduce the weights of channels containing anomalies. The refined output is multiplied with the feature maps from the convolutional structures, achieving multidimensional anomaly suppression across channel, spatial, and scale levels. This produces a final feature map that accurately represents background information, effectively masks anomalies, and provides richer supplementary information for anomaly suppression during decoding.

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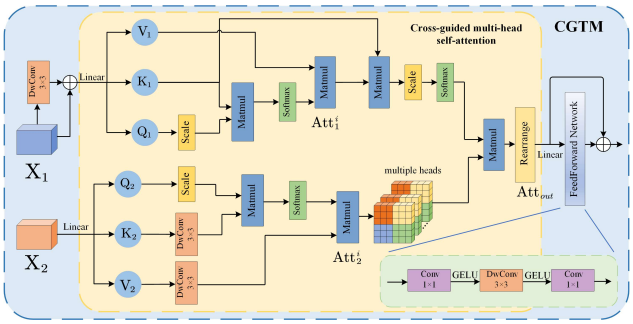


Fig. 4. Schematic of CGTM.

D. Cross-Guided Transformer Module

Transformer models can strengthen features through global attention. Although it performs well in capturing overall data information, it still has limitations, such as insensitivity to modeling local details and loss of complex background details. To tackle these issues and enhance the model's ability to distinguish between background and anomalies during training, we introduce a new transformer CGTM in Fig. 4.

First, we apply the conditional position embedding (CPE) [54] to the lower layer decoder input through a 3×3 DwConv layer. Then, we divide the entire module into a decoding multihead self-attention branch and an encoding multihead self-attention branch. The input lower layer decoder information $\mathbf{X}_1$ and encoder information $\mathbf{X}_2$ are linearly project to obtain the queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$

$$\mathbf{Q}_i = \mathbf{X}_i \mathbf{W}_i^q, \ \mathbf{K}_i = \mathbf{X}_i \mathbf{W}_i^k, \ \mathbf{V}_i = \mathbf{X}_i \mathbf{W}_i^v \tag{6}$$

where $\mathbf{W}_i^q$, $\mathbf{W}_i^k$, and $\mathbf{W}_i^v$ represent the projection weights for the $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$, respectively.

After the mapping is completed, the decoding multihead attention branch performs background smoothing recovery by capturing the spectral dimensional deep features of the decoding information. The obtained $\mathbf{Q}_1$, $\mathbf{K}_1$, and $\mathbf{V}_1$ is processed using nonoverlapping windows (with a size of M×M) to yield $[\tilde{\mathbf{Q}}_1, \tilde{\mathbf{K}}_1, \tilde{\mathbf{V}}_1] \in \mathbb{R}^{H'W'/M \times M^2 \times C'}$. Subsequently, they are divided into multiple heads along the channel dimension, as follows:

$$\begin{aligned} \tilde{\mathbf{Q}}_1 &= \left[\tilde{\mathbf{Q}}_1^1, \tilde{\mathbf{Q}}_1^2, \ldots, \tilde{\mathbf{Q}}_1^h\right] \\ \tilde{\mathbf{K}}_1 &= \left[\tilde{\mathbf{K}}_1^1, \tilde{\mathbf{K}}_1^2, \ldots, \tilde{\mathbf{K}}_1^h\right] \\ \tilde{\mathbf{V}}_1 &= \left[\tilde{\mathbf{V}}_1^1, \tilde{\mathbf{V}}_1^2, \ldots, \tilde{\mathbf{V}}_1^h\right] \end{aligned} \tag{7}$$

where h represents the number of divided heads. During this process, different attention heads can focus on the correlations between different spectral bands, allowing for fine background representation in the spectral dimension. Then, the multihead self-attention mechanism is computed for each head, ensuring that the model can more accurately describe the local details in the background

$$\mathbf{Att}_1^i = \text{Softmax}\left(\frac{\tilde{\mathbf{Q}}_1^i \left(\tilde{\mathbf{K}}_1^i\right)^T}{\sqrt{d}}\right) \tilde{\mathbf{V}}_1^i. \tag{8}$$

In the encoding multihead attention branch, we focus more on the fine-grained background reconstruction of the encoded information. Before the nonoverlapping window division, $\mathbf{K}_2$ and $\mathbf{V}_2$ each undergo a 3×3 DwConv to yield $\tilde{\mathbf{K}}_2 = \text{DwConv}_{3\times3}(\mathbf{K}_2)$ and $\tilde{\mathbf{V}}_2 = \text{DwConv}_{3\times3}(\mathbf{V}_2)$. Compared to the decoding attention branch, we employ two DwConvs to enhance the extraction of local abnormal features, and the subsequent operations are the same. Therefore, the computation for the encoding multihead self-attention branch is as follows:

$$\mathbf{Att}_2^i = \text{Softmax}\left(\frac{\tilde{\mathbf{Q}}_2^i \left(\tilde{\mathbf{K}}_2^i\right)^T}{\sqrt{d}}\right) \tilde{\mathbf{V}}_2^i. \tag{9}$$

To apply greater weight to the decoupled background information, we establish a connection between the features of the encoder and decoder. The attention output from the decoding branch is used as the $\mathbf{Q}$, while the $\mathbf{K}$ are derived from the window-partitioned and $\mathbf{V}$ of the encoding branch's attention. The second combination of $\mathbf{K}_1$ and $\mathbf{Att}_1$ further cross-fuses the related information with the encoded feature values $\mathbf{Att}_2$. This leads to a cross-guided attention computation to restore background details, followed by rearrangement and linear layers to integrate the multiple heads. The final attention computation is as follows:

$$\mathbf{Att}_{\text{out}} = \text{Linear}\left(\text{Re}\left(\text{Softmax}\left(\frac{\mathbf{Att}_1^i \left(\tilde{\mathbf{K}}_1^i\right)^T}{\sqrt{d}}\right)\mathbf{Att}_2^i\right)\right) \tag{10}$$

where $\text{Re}(\cdot)$ indicates the rearrangement operation on the tensor. Finally, $\mathbf{Att}_{\text{out}} \in \mathbb{R}^{H' \times W' \times C'}$ is passed through a feedforward network.

CGTM jointly processes the inputs from the encoder and decoder, leveraging the information exchange between them to enhance the background confidence and weaken the local spatial dependencies of anomalies.

E. SYO Loss

During iterative training, our goal is to align the background pixels in the generated image with those in the original HSI while suppressing the reconstruction of anomalous pixels. Inspired by GAED [38], a background-priority mapping SYO loss function is adopted for HAD, calculated as follows:

$$d_n = \sum_{c=1}^{C} \left[\boldsymbol{t}_{i,j}(c) - \boldsymbol{u}_n(c)\right]^2 \tag{11}$$

where $(n = 1, 2, \ldots, 8)$ represents the number of subvolumes, and $\boldsymbol{t}_{i,j}(c)$ and $\boldsymbol{u}_n(c)$ denote the pixel values of the cth band at position (i, j) for the HSI and the average vector of each subvolume, respectively. The spectral similarity score is computed to update the pixel values and obtain the background-guided map $\mathbf{G}$, as follows:

$$g_{x,y} = \max_n \left(e^{-d_n^2}\right). \tag{12}$$

Therefore, the reconstruction loss in the spatial–spectral domain is defined as follows:

$$\mathcal{L}_m + \mathcal{L}_s = \text{mse}\left(\mathbf{X}, \mathbf{G}\hat{\mathbf{X}}\right) + \text{SmoothL1}\left(\mathbf{X}, \mathbf{G}\hat{\mathbf{X}}'\right) \tag{13}$$

where $\mathbf{G}\hat{\mathbf{X}}$ is the Hadamard product result of $\mathbf{G}$ and $\hat{\mathbf{X}}$.

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The background-guided map assigns higher confidence to background pixels and lower weights to anomalies, reducing their influence in the loss function and suppressing anomalies. The mse loss function preserves HSI's spatial continuity and spectral gradient characteristics of object edges, while the SmoothL1 loss function is sensitive to large gradient changes (e.g., anomalies) and robust to small changes (e.g., textures), avoiding excessive punishment on edge details. To evaluate reconstruction quality, we combine mse loss and SmoothL1 loss to balance anomaly suppression and the robustness of background reconstruction in the task.

In practical image reconstruction tasks, important anomalies details are often concentrated in the high-frequency components of the frequency domain. However, loss functions such as mse and SmoothL1 tend to overlook these features in the frequency domain, leading to the reconstruction of anomalies. To solve this, we propose a new frequency-domain loss [55] that transforms the image into the frequency domain and compares distribution differences, improving sensitivity to high-frequency anomalies. We combine mse, SmoothL1, and frequency-domain losses to optimize background reconstruction and overcome the limitations of relying only on high-frequency suppression. The SYO suppresses anomalies in both spatial and frequency domains, while the losses also optimize and preserve edge and texture details. The frequency-domain representation of the input image is obtained using a discrete Fourier transform (DFT), as follows:

$$\mathbf{X}(u,v) = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} \mathbf{x}(i,j)\cdot e^{-i2\pi\left(\frac{ui}{H}+\frac{vj}{W}\right)}. \tag{14}$$

The frequency mask is applied for separation, where frequencies greater than the default distance threshold of 0.5 represent high-frequency components with rapid spatial changes, corresponding to anomalies

$$M(f) = \begin{cases} 1, & \text{if } \|f\| > \text{threshold} \\ 0, & \text{if } \|f\| \le \text{threshold} \end{cases} \tag{15}$$

$$\mathbf{X}_{\text{low}} = \mathbf{X}(u,v)\cdot\left(\bar{M}(f)\right), \mathbf{X}_{\text{high}} = \mathbf{X}(u,v)\cdot M(f) \tag{16}$$

where $\|f\| = (f_u^2 + f_v^2)^{1/2}$ is the Euclidean distance of $f(u,v)$ and $\bar{M}(f) = 1 - M(f)$. The final frequency domain loss is defined as a weighted sum of the two components

$$\mathcal{L}_f = \frac{1}{\text{HW}}\left\|(1-\omega)\left(\mathbf{X}_{\text{low}} - \hat{\mathbf{X}}_{\text{low}}\right) + \omega\left(\mathbf{X}_{\text{high}} - \hat{\mathbf{X}}_{\text{high}}\right)\right\|_F^2 \tag{17}$$

where ω is the weighting parameter that assigns a lower weight to the high-frequency component to enhance anomaly suppression. By using the above designs, a larger weight is assigned to the background areas as much as possible, thereby increasing their influence in the loss calculation. The overall loss function is a joint constraint considering both spatial–spectral and frequency domains as follows:

$$\mathcal{L} = \mathcal{L}_m + \mathcal{L}_s + \mathcal{L}_f. \tag{18}$$

Algorithm 1 D³Former

Input: 1) The observed HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$; 2) uniform noise $\mathbf{X}_u \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times 4N}$.

Output: Final detection result $\mathbf{R}$.

Stage 1: Training Phase;

1: $\mathbf{X}$ is processed by the FF encoder to extract abstract background feature $\mathbf{E}_{i,j}$.
2: ARMAB calculates the residual $\mathbf{S}$ using Eq. (3) and performs anomaly masking on $\mathbf{S}$ to obtain a pure coarse background information $\mathbf{B}$.
3: $\mathbf{E}_{i,j}$ and $\mathbf{X}_u$ are restored into background features through the FF decoder and BPD.
4: During the decoding phase, $\mathbf{B}$ is simultaneously delivered to BPD as supplementary information.
5: The input data and pure background are used as training targets for the reconstruction network D³Former.
6: Minimize SYO loss function guides the model to suppress anomalies.
7: Integrate background decoding feature $\mathbf{D}_{i,j}$ from each stage through BFFM.
8: Obtain the final reconstructed background HSI_1.

Stage 2: Detection Phase;

9: Compute the reconstruction residual for each pixel in $\mathbf{X}$ by Eq. (19).

F. BFFM and Detection Phase

In this BFFM shown in Fig. 5, the output $\mathbf{X}_1$ from the lower layer BPD and the output $\mathbf{X}_3$ from the upper layer FF decoder are first concatenated along the channel dimension. The fused feature map is then processed through a spatial attention mechanism to enhance information interaction between different spatial locations, highlighting the key information in the background. $\mathbf{X}_1$ and $\mathbf{X}_3$ are processed through depth attention to capture important fine-grained background details.

Next, the output of the spatial attention is added to the result of $\mathbf{X}_3$ after depth attention processing. This summed result is then element-wise multiplied with the feature map of $\mathbf{X}_1$ after depth attention refining, enhancing important background information while suppressing anomalous interference, generating the final output feature. The entire process maintains background integrity through the collaborative effect of depth attention and spatial attention, ultimately producing a clean background $\hat{\mathbf{X}}$.

Finally, the reconstruction error is calculated to represent the detection result as the following equation:

$$\mathbf{R}_{i,j} = \left\|\mathbf{X}_{(i,j,:)} - \hat{\mathbf{X}}_{(i,j,:)}\right\|_2^2 \tag{19}$$

where $(\cdot)_{i,j,:}$ is the spectral vector at position (i, j). The detailed steps of D³Former are introduced in Algorithm 1.

III. Experiment Result and Analysis

In the experiments, we validated the performance of the D³Former model on six real datasets. The D³Former model was implemented in Python (version 3.10.6) using the PyTorch framework with PyCharm 2021.3 as the compiler. The computational environment was a computer equipped with an

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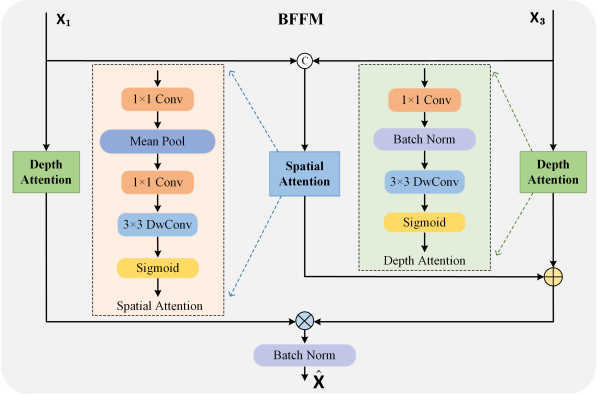


Fig. 5. Schematic representation of BFFM.

NVIDIA 3090 24-GB GPU. The datasets, shown in Fig. 6, include their pseudo-color images and corresponding ground truth (GT) values.

A. Dataset Description

1) Pavia Dataset [33]: This real dataset was obtained from an airborne sensor called the Reflective Optics System Imaging Spectrometer (ROSIS), with a spatial resolution of 1.3 m and a wavelength range of 430–860 nm. The image has a size of 100 × 100 with 102 spectral bands and contains 71 anomalies.

2) San Diego (SD) Dataset [56]: This HSI was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in San Diego, CA, USA, with a spatial resolution of 3.5 m and a wavelength range of 370–2510 nm. The spatial dimensions are 100 × 100, with 189 spectral bands, excluding low signal-to-noise ratio (SNR) and absorption bands. Referred to as SD, this HSI identifies 58 airport pixels as anomalies.

3) Texas Coast (TC) Dataset [57]: The two city datasets were captured by the AVIRIS sensor over the coast of Texas in the USA in 2010. Both of these scenes have a spatial resolution of 17.2 m and a size of 100 × 100 pixels. The first scene contains 67 anomalous pixels and 204 spectral bands, with a wavelength range of 450–1350 nm. Buildings of different scales are considered anomalies. The second scene has 207 bands, and all bands were used in each experiment. These two HSIs are referred to as TC-1 and TC-2.

4) Gainesville (GA) Dataset [57]: This dataset represents an urban scene captured by the AVIRIS airborne sensor in Gainesville, FL, USA. By removing water-absorbing and SNR bands from the original data, 191 bands were retained for experimentation. In addition, the dataset has a spatial resolution of 3.5 m and a size of 100 × 100 pixels. This dataset contains 156 anomalous pixels, accounting for 1.56% of the entire image, which are yet to be explored.

5) Salinas Dataset [47]: The synthetic image was derived from the AVIRIS sensor and selected a background subimage from the Salinas scene. The anomalous target insertion method was applied after using a linear mixing model and is written as follows:

$$\mathbf{n} = f_a\mathbf{a} + (1 - f_a)\,\mathbf{b} \tag{20}$$

where $\mathbf{a}$, $\mathbf{b}$, $\mathbf{n} \in \mathbb{R}^{L\times 1}$ represent the spectral vectors of the anomaly, background, and final mixture, respectively, while f_a denotes the abundance fraction of $\mathbf{a}$. 25 targets were randomly embedded in the subimage, with corresponding abundance f_a values ranging from 0.04 to 1.

B. Experimental Settings

We selected ten methods for comparison, including GRX [9], CRD [22], FrFE [12], RCRD [23], PCA-TLRSR [58], LEBSR [26], GAED [38], Auto-AD [36], PDBSNet [46], and MSNet [45]. GRX is a method based on probability distribution. CRD, RCRD, and PCA-TLRSR are representation-based methods. FrFE employs frequency-domain transformations to detect anomalies. LEBSR detects anomalies by learning the background end members. GAED, Auto-AD, PDBSNet, and MSNet reconstruct the background in a self-supervised manner. For our method, the learning rate is set to 10^{-4}. Training epochs and the regularization parameters ω were set to 800, 600, 800, 600, and 1400, 0.1, 0.3, 0.3, 0.1, and 0.1 for the Pavia, SD, TC, GA, and Salinas datasets, respectively. We suggest setting the training epochs within the range of [600, 1000] to achieve more promising and stable learning performance.

To measure the detection performance of different AD methods, the receiver operating characteristic (ROC) curve is widely used for intuitive observation. We use the ROC curves of (P_f, P_d) and (P_f, τ) to represent overall detection accuracy and background suppression capability, respectively. P_f and P_d are related to τ, and they can be expressed as follows:

$$P_d^{(\tau)} = \frac{TP^{(\tau)}}{N_{\text{ano}}}, P_f^{(\tau)} = \frac{FP^{(\tau)}}{N_{\text{bkg}}} \tag{21}$$

where $\text{TP}^{(\tau)}$ and $\text{FP}^{(\tau)}$ denote the number of anomalies and background pixels whose detection values are greater than a given threshold τ, and N_{ano} and N_{bkg} are the total number of anomalies and background pixels.

C. Experimental Results and Analysis

1) Quantitative Comparisons: Table I shows the area under the curve (AUC) values of the ROC curves, where $\text{AUC}(P_f, P_d)$ indicates detection accuracy, while $\text{AUC}(P_f, \tau)$ measures background suppression capability. From the table, it can be seen that our proposed D^{3}Former method achieves the highest score for $\text{AUC}(P_f, P_d)$ across six datasets. Regarding the $\text{AUC}(P_f, \tau)$ for this method, it scored the lowest on five datasets, while ranking second on the remaining dataset. Overall, D^{3}Former achieved the best average scores in $\text{AUC}(P_f, P_d)$ and $\text{AUC}(P_f, \tau)$ across all six datasets, owing to its effective reconstruction of the background and suppression of anomalies through the dual-decoder transformer network structure, which in turn augments detection rates and background suppression effectiveness.

2) Visual Comparisons: To visually reflect the detection results, we present the detection outcomes of different AD algorithms on six HSI datasets, as shown in Fig. 7. From Fig. 7, it is evident that the RX algorithm misses many anomalous targets across these six datasets while misclassifying a

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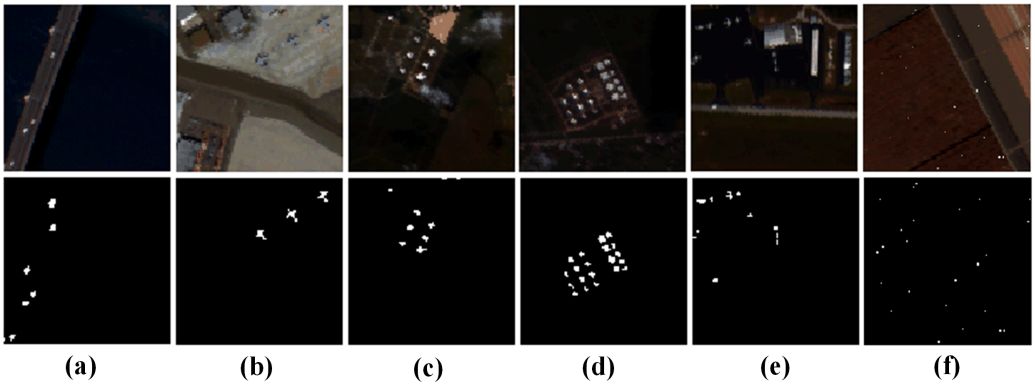


Fig. 6. Pseudo-color image and GT of experimental datasets. (a) Pavia, (b) SD, (c) TC-1, (d) TC-2, (e) GA, and (f) Salinas.

TABLE I
AUC VALUES OF ALL COMPARED METHODS ON DIFFERENT DATASETS

Dataset	AUC values $AUC(P_d,\ P_f)$										
	GRX 1990	CRD 2015	FrFE 2019	RCRD 2022	PCA-TLRSR 2023	LEBSR 2024	GAED 2022	Auto-AD 2022	PDBSNet 2023	MSNet 2024	Ours
Pavia	0.9934	0.9570	0.9935	0.9922	0.9921	0.9946	0.9948	0.9942	0.9968	0.9954	**0.9992**
San Diego	0.8885	0.9493	0.9788	0.9267	0.9908	0.9871	0.9890	0.9513	0.9858	0.9877	**0.9931**
Texas-Coast-1	0.9907	0.9740	0.9815	0.9845	0.9711	0.9882	0.9441	0.9816	0.9869	0.9924	**0.9952**
Texas-Coast-2	0.9946	0.9183	0.9822	0.9717	0.9701	0.9961	0.9959	0.9903	0.9954	0.9938	**0.9991**
Gainesville	0.9513	0.9469	0.9626	0.9759	0.9728	0.9535	0.9726	0.9516	0.9711	0.9631	**0.9837**
Salinas	0.8073	0.9653	0.9524	0.9095	0.9531	0.9720	0.9683	0.9871	0.9898	0.9822	**0.9932**
Average	0.9376	0.9518	0.9752	0.9601	0.9750	0.9819	0.9775	0.9760	0.9876	0.9858	**0.9939**

Dataset	AUC values $AUC(P_f,\ \tau)$										
	GRX 1990	CRD 2015	FrFE 2019	RCRD 2022	PCA-TLRSR 2023	LEBSR 2024	GAED 2022	Auto-AD 2022	PDBSNet 2023	MSNet 2024	Ours
Pavia	0.0258	0.0927	0.0276	0.0121	0.0980	0.0012	0.0056	0.0011	0.0016	0.0014	**0.0008**
San Diego	0.0381	0.0448	0.0194	0.1651	0.1123	0.0066	0.0100	0.0092	0.0098	0.0069	**0.0055**
Texas-Coast-1	0.0556	0.0267	0.0630	0.0944	0.2035	0.0177	0.0572	0.0159	0.0134	0.0061	**0.0058**
Texas-Coast-2	0.0135	0.0124	0.0161	0.0162	0.0470	0.0047	0.0045	0.0022	0.0007	**0.0005**	0.0009
Gainesville	0.0351	0.0275	0.0378	0.0801	0.1173	0.0466	0.0168	0.0316	0.0172	0.0162	**0.0100**
Salinas	0.0314	0.0262	0.0412	0.0890	0.1011	0.0138	0.0773	0.0022	0.0018	0.0025	**0.0011**
Average	0.0333	0.0384	0.0342	0.0762	0.1132	0.0151	0.0286	0.0104	0.0074	0.0056	**0.0040**

large number of background pixels. This is because RX is based on simple statistical rules, which do not adapt well to the complex statistical distributions of HSI data. Although the CRD and RCRD methods present a relatively complete background, this leads to difficulties in distinguishing anomalies from the background. FrFE and PCA-TLRSR also produce many erroneous anomaly results during detection. While LEBSR and Auto-AD are able to detect anomalous targets across all datasets, the background is still quite pronounced. In contrast, the GAED, PDBSNet, and MSNet methods effectively detect anomalies, with MSNet performing better in terms of background suppression. Compared to the other methods, the proposed D³Former method not only successfully detects anomalous targets but also effectively separates anomalies from the background, significantly improving detection accuracy and achieving excellent background suppression.

To further qualitatively analyze the detection performance of different methods, we present two ROC curves in Fig. 8. For better visualization, they are displayed on a logarithmic scale. The ROC curves of (P_f, P_d) and (P_f, τ) are close to the upper left and lower left corners, respectively, indicating that the detector can achieve a high probability of detection with low false alarms. From the ROC curve of (P_f, P_d), D³Former shows a higher detection probability across selected datasets compared to other methods at various false alarm rates. Overall, the proposed D³Former method demonstrates

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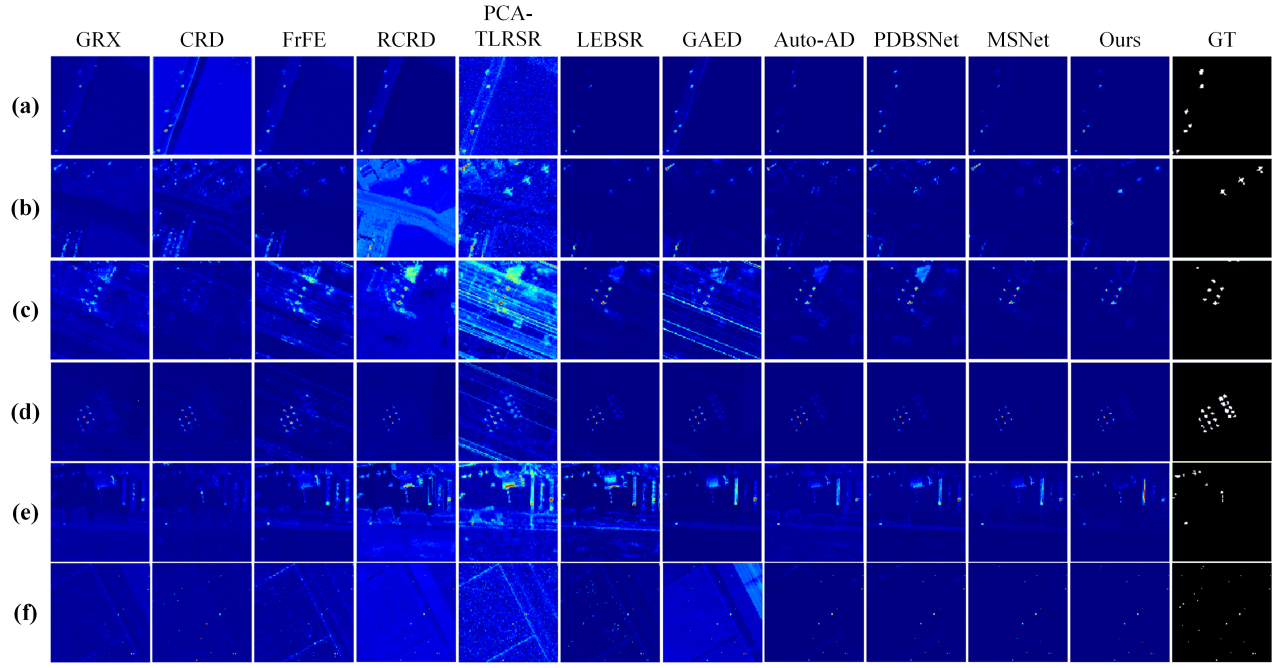


Fig. 7. Detection maps of different methods. (a) Pavia, (b) SD, (c) TC-1, (d) TC-2, (e) GA, and (f) Salinas.

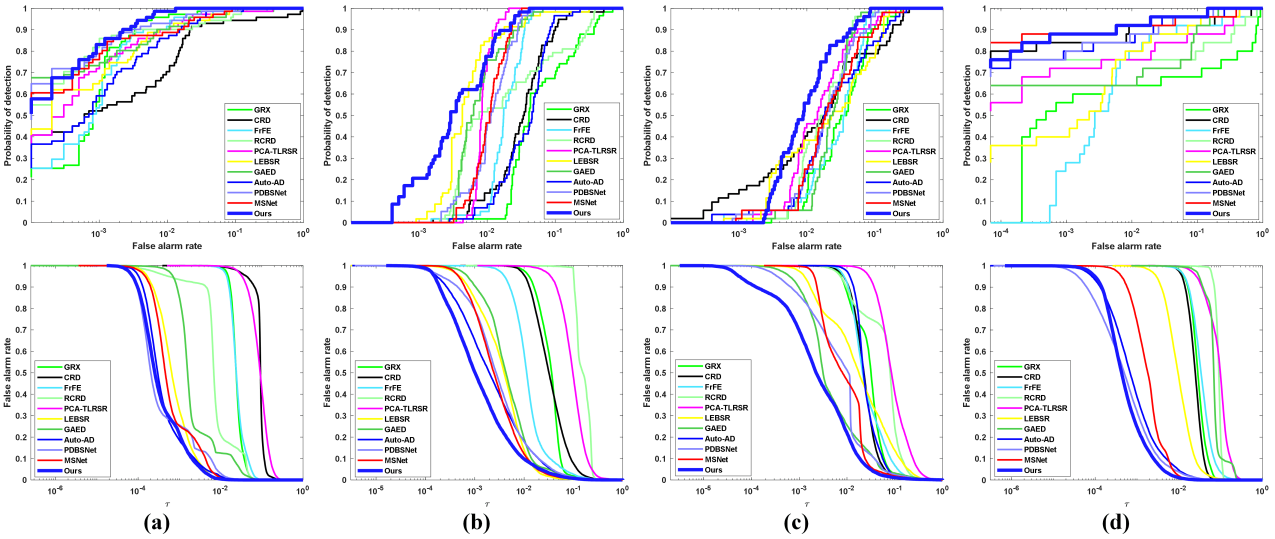


Fig. 8. ROC curves of (P_f, P_d) and (P_f, τ) for different methods on (a) Pavia, (b) SD, (c) GA, and (d) Salinas datasets.

superior performance on the ROC curve (P_f, P_d) compared to other methods, while its performance on the ROC curve (P_f, τ) is lower than that of other methods. This indicates that our method can maintain a high detection rate and a low false positive rate across all datasets, thereby exhibiting consistency in visual detection results, as shown in Fig. 8.

3) Separability Analysis: To evaluate the separation capability between the background and anomalies, Fig. 9 presents box plots based on the detection values, revealing the separability of anomalous targets from background pixels, which is used to measure detection performance. The red box plot represents anomalous targets, while the blue box plot represents the background. Overall, different methods exhibit varying separation effects across the datasets. A larger boundary between the background and anomalies indicates that anomalous pixels can be more easily detected, suggesting better detection performance of the method. In datasets such as SD and TC-1, the anomalies from the RCRD and GAED methods significantly overlap with the background, indicating their weak separation capabilities. A small background box suggests that the background is more concentrated, demonstrating a strong background suppression effect. However, RCRD and PCA-TLRSR exhibit a highly divergent background across various datasets, showing significant limitations in background suppression. In contrast, D^{3}Former shows almost no overlap between the background and anomalies across the six datasets,

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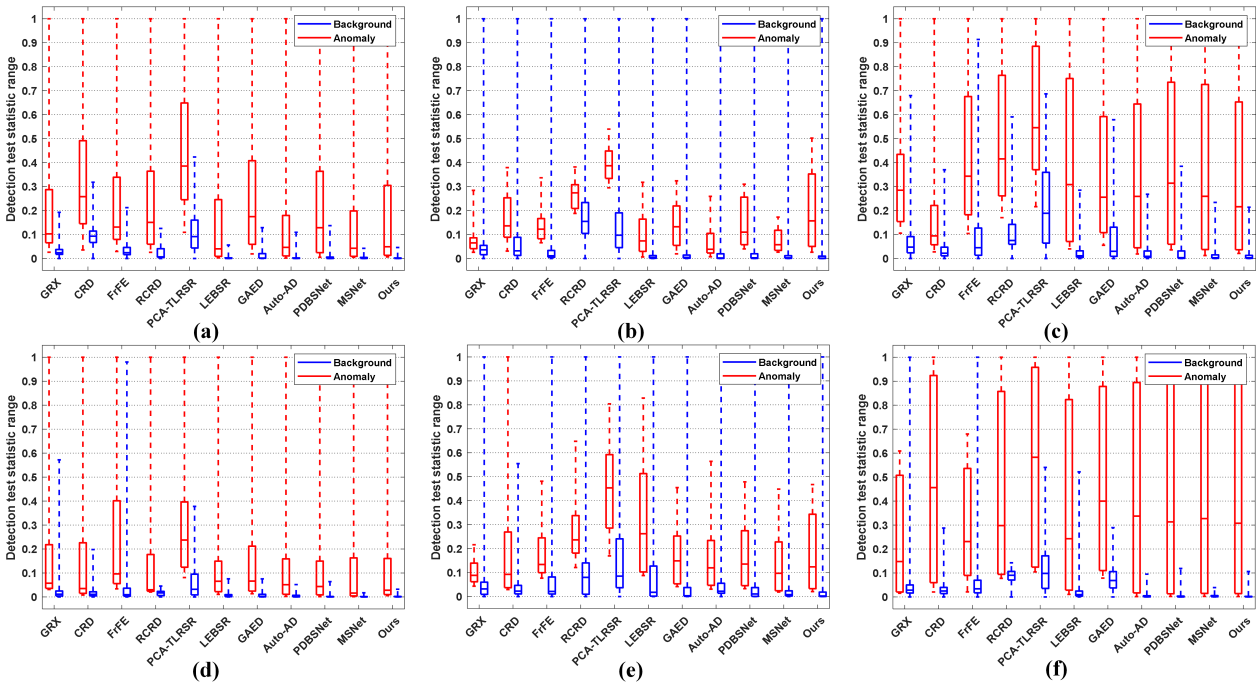


Fig. 9. Background-anomaly separability results for different methods on (a) Pavia, (b) SD, (c) TC-1, (d) TC-2, (e) GA, and (f) Salinas datasets.

with the background appearing highly concentrated. These results demonstrate that the method has significant advantages in background suppression and anomaly separation.

D. Validation of Network Structure

We first effectively simulate complex scenarios in HAD by altering the input images of the BPD in the dual-decoder. In addition, we introduce different network variants to conduct extensive experiments on classic datasets to validate the effectiveness of our proposed D^{3}Former.

1) Distinct Inputs for BPD: We input two different images into the encoder and BPD. One input is the original image, while the other input is a different noise image. This effectively simulates complex scenarios in HAD and enhances the robustness of the model. To demonstrate the effectiveness of the noise modeling approach, we introduce three different noise inputs for validation.

1) ***w/ Gaussian:*** In this variant, the input to the encoder remains unchanged, while the input to the lower layer BPD is transformed into an image of Gaussian noise.
2) ***w/ salt-pepper:*** In this variant, the input to the encoder remains unchanged, while the input to the BPD is transformed into an image of salt-and-pepper noise.
3) ***w/ striped:*** In this variant, the input to the encoder remains unchanged, while the input to the BPD is transformed into an image of stripe noise.

Table II shows the impact of using different noise inputs on detection accuracy. The results indicate that the use of uniform noise leads to better detection performance. The uniform noise is evenly distributed without prominent patterns, preventing the model from extracting abnormal spatial–spectral features. At the same time, it can simulate the uniform statistical distribution of background, serving as a foundation for the model to reconstruct the estimated pure background in combination with the structured features. The introduction of uniform noise mainly prevents the model from learning anomalies and preliminarily establishes the background statistical distribution. In contrast, the Gaussian noise follows a specific statistical distribution, which will guide the model to fit Gaussian characteristics and may mistakenly learn anomalies. Salt-pepper noise introduces extreme pixel-level values, increasing the risk of false alarms. Meanwhile, the strong directional structure of striped noise will interfere with background learning.

TABLE II
ABLATION STUDY OF DUAL-DECODER WITH DIFFERENT NOISE CONDITIONS

AUC Values	Model	Pavia	SD	TC-2	GA
$AUC(P_d,\ P_f)$	w/gaussian	0.9988	0.9923	0.9978	0.9796
	w/salt-pepper	0.9982	0.9927	0.9970	0.9826
	w/striped	0.9978	0.9920	0.9974	0.9812
	D^{3}Former (Ours)	**0.9992**	**0.9931**	**0.9991**	**0.9837**

2) Dual-Decoder: We employ a dual-decoder structure with different inputs to extract more meaningful supplementary information, enriching the communication of contextual information and enabling strong constraints on anomalies during background reconstruction. To demonstrate the effectiveness of our dual-decoder, we introduce four different network variants under mse for validation.

1) ***w/o BPD and ARMAB:*** Variant 1, we directly remove the entire BPD part, retaining the FF encoder–decoder components in the network.

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TABLE III
ABLATION STUDY OF DUAL-DECODER STRUCTURE WITH DIFFERENT NETWORK MODULES

AUC Values	Model	Pavia	San Diego	TC-1	TC-2	Gainesville	Salinas	Average
$AUC(P_d,\ P_f)$	D^3Former w/o BPD and ARMAB	0.9676	0.9600	0.9725	0.9753	0.9543	0.9696	0.9666
	D^3Former w/ BFFM	0.9688	0.9613	0.9731	0.9766	0.9551	0.9701	0.9675
	D^3Former w/ CNN decoder	0.9761	0.9737	0.9799	0.9825	0.9641	0.9767	0.9755
	D^3Former w/ CNN and ARMAB	0.9879	0.9811	0.9842	0.9894	0.9754	0.9824	0.9834
	D^3Former (Ours)	**0.9952**	**0.9892**	**0.9902**	**0.9962**	**0.9784**	**0.9891**	**0.9897**

TABLE IV
ABLATION EXPERIMENTS ON SIX DATASETS

AUC Values	ARMAB	CGTM	SYO Loss	Pavia	San Diego	TC-1	TC-2	Gainesville	Salinas	Average
$AUC(P_d,\ P_f)$				0.9688	0.9613	0.9731	0.9766	0.9551	0.9701	0.9675
	✓			0.9838	0.9774	0.9826	0.9885	0.9692	0.9812	0.9805
		✓		0.9777	0.9738	0.9788	0.9881	0.9701	0.9766	0.9775
			✓	0.9721	0.9716	0.9755	0.9820	0.9633	0.9751	0.9732
	✓	✓		0.9952	0.9892	0.9902	0.9962	0.9784	0.9891	0.9897
	✓		✓	0.9911	0.9849	0.9878	0.9917	0.9756	0.9865	0.9863
	✓	✓	✓	**0.9992**	**0.9931**	**0.9952**	**0.9991**	**0.9837**	**0.9932**	**0.9939**
$AUC(P_f,\ \tau)$				0.0091	0.0155	0.0132	0.0100	0.0197	0.0092	0.0128
	✓			0.0043	0.0108	0.0087	0.0054	0.0145	0.0061	0.0083
		✓		0.0040	0.0131	0.0089	0.0068	0.0188	0.0066	0.0097
			✓	0.0074	0.0133	0.0112	0.0077	0.0190	0.0078	0.0110
	✓	✓		0.0015	0.0076	0.0066	0.0020	0.0112	0.0022	0.0052
	✓		✓	0.0022	0.0089	0.0079	0.0038	0.0159	0.0034	0.0070
	✓	✓	✓	**0.0008**	**0.0055**	**0.0058**	**0.0009**	**0.0100**	**0.0011**	**0.0040**

2) ***w/BFFM:*** Variant 2, we directly remove the entire BPD portion, retaining the FF encoder–decoder components and BFFM.
3) ***w/CNN:*** Variant 3, we modify the BPD structure by adopting a standard convolutional framework, composed of convolutional layers, batch normalization, and LeakyReLU.
4) ***w/CNN and ARMAB:*** Variant 4, we retain the use of the CNN framework for BPD and ARMAB.

Table III presents the results of the ablation study on the dual-decoder. When removing the entire lower layer BPD, and leaving only the FF encoder and decoder, the AUC(P_d, P_f) value significantly decreases across the datasets. This suggests the dual-decoder structure captures more useful information, improving feature reconstruction. Adding BFFM alone has little effect on AUC(P_d, P_f), indicating that the role of the dual-decoder architecture is to serve as a bridge for introducing diverse feature processing for ARMAB and CGTM, enabling more refined background recovery. In comparison, using the traditional CNN framework for the BPD results in a decrease in the AUC(P_d, P_f) values by 1.91%, 1.55%, 1.03%, 1.37%, 1.43%, and 1.24% on the six datasets. This demonstrates that our designed dual-transformer structure in CGTM better integrates features at different levels and effectively mitigates the influence of anomalous targets during background fusion. Furthermore, when adding ARMAB to the CNN framework, a significant improvement in the AUC(P_d, P_f) value is achieved, further validating that ARMAB can significantly help recover lost edge and texture details.

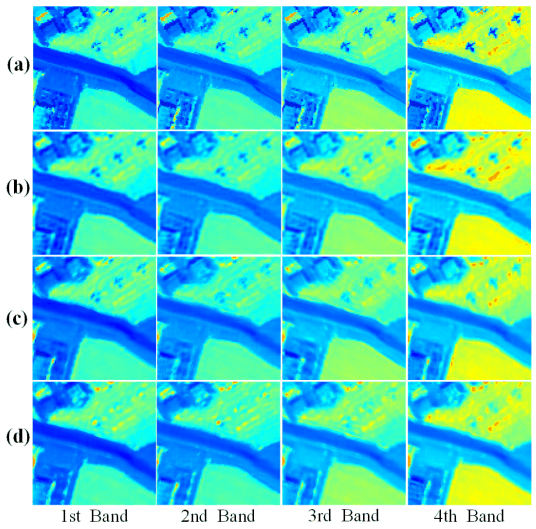


Fig. 10. Output map of the reconstruction results on the SD scene. (a) Original HSI, (b) remove the BPD, (c) only use CNN, and (d) ours.

Fig. 10 shows the background reconstruction results for the two variants and our D^3Former on the SD scene across four different spectral bands. When the BPD is removed, the generated background image in Fig. 10(b) prominently reconstructs the aircraft as an anomalous target, and the background appears quite blurry. This emphasizes the importance of our dual-decoder in preserving the spatial correlation of the background. In Fig. 10(c), although using a traditional CNN as the BPD shows a trend of suppressing anomalies,

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the anomalies are still inevitably reconstructed. In contrast, the CGTM in the dual-decoder, as illustrated in (d), can clearly eliminate anomalies, ensuring a cleaner anomaly-free background.

E. Ablation Study

1) Specific Module Ablation: To further illustrate the rationality and effectiveness of the design of each module, we use a dual-decoder network that only includes SWTM and BFFM as the baseline for D³Former. The proposed D³Former method mainly includes ARMAB, CGTM, and SYO loss, and we analyze the impact of each component for the model. Table IV shows the results of the overall ablation experimental results. We observed that using CGTM alone can enhance the model's background representation capability, which led to higher AUC(P_d, P_f) values. The introduction of the SYO loss function, as a regularization mechanism in the frequency domain, can further suppress anomalies. Upon comparison, adding ARMAB alone can significantly improve detection accuracy for each dataset, indicating that ARMAB is effective in suppressing anomalies as well as pure background reconstruction. This is because ARMAB can explicitly model the anomaly masking mechanism and promote the model to filter out anomalies. The combination of ARMAB and CGTM results in an average AUC(P_d, P_f) value improvement of 2.22%, as CGTM can weaken the local spatial dependencies of anomalies through cross-layer attention, making it harder for anomalous features to be retained during multiscale feature fusion. In summary, the core of our method is to combine the advantages of these modules through collaborative ensemble optimization. As a result, our D³Former achieves the highest AUC(P_d, P_f) value and the lowest AUC(P_f, τ) value.

2) Different Loss Functions: To investigate the benefits of different loss functions, we have compared our SYO with five combinations of loss functions.

1) ***w/mse:*** Combination 1, we optimize HSI_1 only using the mse loss function.
2) ***w/$\mathcal{L}_f$:*** Combination 2, we optimize HSI_1 only using the frequency-domain loss function $\mathcal{L}_f$.
3) ***w/mse+mse:*** Combination 3, we simultaneously constrain both HSI_1 and HSI_2 using mse loss function.
4) ***w/mse+SmoothL1:*** Combination 4, we simultaneously constrain both HSI_1 and HSI_2 using mse and SmoothL1 loss functions.
5) ***w/mse+mse+$\mathcal{L}_f$:*** Combination 5, we simultaneously constrain both HSI_1 and HSI_2 using mse and $\mathcal{L}_f$ loss functions.

Table V reports the AUC performance for different combinations of loss functions. It can be observed that using the frequency-domain loss $\mathcal{L}_f$ alone shows good performance in improving the AUC (P_d, P_f) value. This suggests that $\mathcal{L}_f$ can suppress the high-frequency components dominated by anomalies, while having a relatively small impact on texture detail suppression. Applying the mse loss function to optimize both HSI_1 and HSI_2 can improve the AUC performance, indicating that it plays a positive role in regulating the supplementary background information. Replacing the mse loss function with the SmoothL1 loss function can yield even better performance, confirming that the smoothing property of the SmoothL1 loss function reduces the dominance of anomalous pixels. To further verify the anomaly suppression capability of the frequency-domain loss $\mathcal{L}_f$, synergizing it with mse and SmoothL1 achieves the best detection performance. To some extent, the introduction of $\mathcal{L}_f$ loss function and the SYO loss function can balance anomaly suppression with texture preservation, significantly improving the model's performance.

TABLE V
ABLATION STUDY OF OUR D³FORMER WITH DIFFERENT COMBINATIONS OF LOSS FUNCTIONS

AUC Values	Loss	Pavia	SD	TC-2	GA
$AUC(P_d,\ P_f)$	w/MSE	0.9952	0.9892	0.9962	0.9784
	w/$\mathcal{L}_f$	0.9969	0.9903	0.9970	0.9809
	w/MSE+MSE	0.9967	0.9915	0.9960	0.9803
	w/MSE+SmoothL1	0.9974	0.9912	0.9977	0.9815
	w/MSE+MSE+$\mathcal{L}_f$	0.9983	0.9922	0.9984	0.9822
	w/SYO (Ours)	**0.9992**	**0.9931**	**0.9991**	**0.9837**

TABLE VI
COMPARED WITH SSL HAD METHODS

AUC Values	Methods	Pavia	SD	TC-2
$AUC(P_d,\ P_f)$	NL2Net	0.9916	0.9885	0.9972
	MSNet	0.9954	0.9877	0.9938
	D³Former (Ours)	**0.9992**	**0.9931**	**0.9991**

F. Relationship With SSL HAD Methods

To illustrate the relationship between our method and SSL HAD methods, a comparison of the performance of the recent SSL methods on three datasets is presented, as shown in Table VI. Overall, the proposed D³Former performs the best. By contrast, the NL2Net [43] focuses on spatial feature modeling, while our D³Former introduces a frequency-domain suppression mechanism, which can suppresses anomalies more comprehensively when reconstructing the background. In addition, although both D³Former and MSNet [45] adopt a multiscale structure, D³Former introduces a dual-decoder structure based on a transformer that considers feature diversity, enabling more refined reconstruction of background features. Compared to other SSL methods such as CSSBR [42] and SAP [41], these new frameworks (such as transfer learning) can be incorporated into our future work. This would further improve the generalization and adaptability of the proposed method, making it better suited for practical engineering applications.

G. Discussion

To verify the originality and versatility of the proposed D³Former, we conduct a comparative analysis using different background priors and applying them to various methods. Although our method borrows the background guidance map G from GAED, it modifies it through an adaptive weighting strategy. As shown in Table VII, comparative experiments

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TABLE VII

PERFORMANCE OF OUR D^{3}FORMER USING DIFFERENT BACKGROUND PRIORS

AUC Values	Priors	Pavia	SD	TC-2	GA
$AUC(P_d,\ P_f)$	w/RX	0.9985	0.9928	0.9988	0.9830
	w/M	0.9988	**0.9833**	0.9989	0.9833
	w/G	**0.9992**	0.9931	**0.9991**	**0.9837**

TABLE VIII

USE OF BACKGROUND PRIORS IN DIFFERENT HAD METHODS

AUC Values	Methods	Pavia	SD	TC-2	GA
$AUC(P_d,\ P_f)$	Auto-AD w/G	0.9950	0.9555	0.9889	0.9531
	GAED w/G	0.9948	0.9890	0.9959	0.9726
	MSNet w/G	0.9942	0.9867	0.9952	0.9645
	D^{3}Former (Ours)	**0.9992**	**0.9931**	**0.9991**	**0.9991**

show that even when replacing the background guidance map G with some other simple and representative background guidance maps, such as the classical RX [9] predetection and background weight map M [36], our model's detection accuracy remains superior, far outperforming most of the comparison methods in Table I. This strongly proves that the benefits of our architecture and modules are the cornerstone of performance.

As shown in Table VIII, we applied the background guidance map G from GAED to different detection frameworks, such as the Auto-AD [36] and MSNet [45], and the results show that the detection accuracy of these detection frameworks did not significantly improve, with only a marginal performance enhancement (compared in Table I). In addition, our method's accuracy consistently exceeds that these detection frameworks equipped with the background guidance map G from GAED, as well as the original GAED. This clearly demonstrates that G is not the key to the superior performance of our D^{3}Former model. Instead, it is the combination of our original network architecture and modules that achieves the two key goals in HAD, i.e., pure background reconstruction and anomaly suppression. As a result, our method outperforms existing SOTA approaches in terms of detection performance.

IV. CONCLUSION

In this article, we propose a novel network called D^{3}Former for HAD. Differing from previous HAD methods, D^{3}Former promotes information flow interaction to model the anomaly-free background by introducing a dual-decoder and dual-transformer structure. Specifically, we design an anomaly feature cleaning auxiliary task during the pure background decoding phase and leverage CGTM to guide the clean reconstruction of the background, further diminishing the representation of anomalies. In addition, by synergistically optimizing the loss function, we achieve suppression of anomalies in both the spatial–spectral and frequency domains. Extensive experimental results demonstrate that D^{3}Former outperforms current state-of-the-art methods on multiple benchmarks.

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